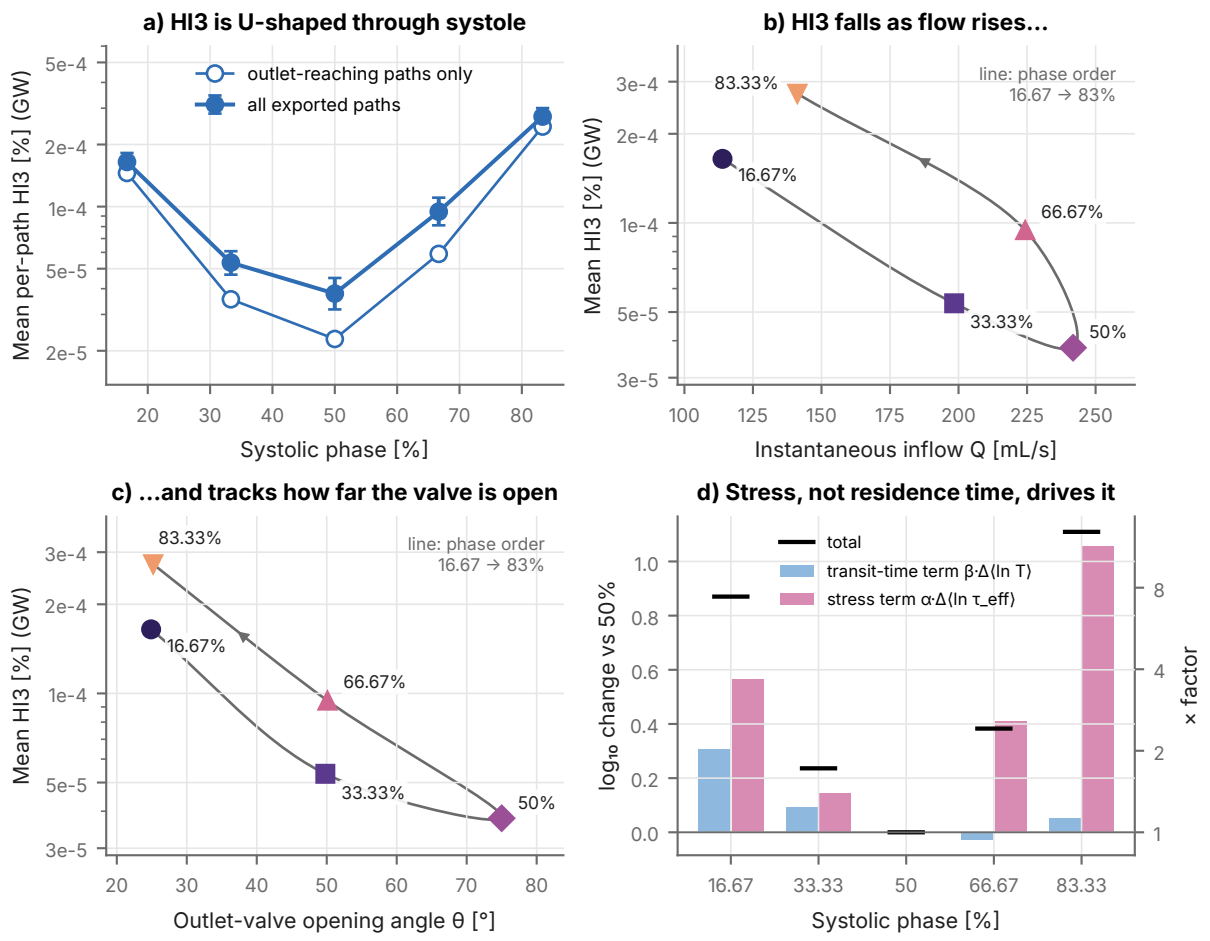
a) HI3 is U-shaped through systole
outlet-reaching paths only
all exported paths
Mean per-path HI3 [%] (GW)
Systolic phase [%]
b) HI3 falls as flow rises…
line: phase order
16.67 → 83%
83.33%
16.67%
66.67%
33.33%
50%
Mean HI3 [%] (GW)
Instantaneous inflow Q [mL/s]
c) …and tracks how far the valve is open
line: phase order
16.67 → 83%
83.33%
16.67%
66.67%
33.33%
50%
Mean HI3 [%] (GW)
Outlet-valve opening angle θ [°]
d) Stress, not residence time, drives it
total
transit-time term β·Δ⟨ln T⟩
stress term α·Δ⟨ln τ_eff⟩
log₁₀ change vs 50%
× factor
16.67
33.33
50
66.67
83.33
Systolic phase [%]